Phase I/II Statistical Process Monitoring SECOM Data

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統計專論 (III) Final Report

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Project Objective

- Dataset: UCI SECOM
- Objective: Construct SPC workflow

$$\text{Phase I} \rightarrow \text{Control chart design} \rightarrow \text{Phase II}$$

- Main comparison: Shewhart, EWMA, CUSUM, and CPD charts
- Final:
  1. Which chart signals first ?
  2. Whether the pattern suggests mean shift, variance shift, or structural change ?

Selected Variable and Preprocessing

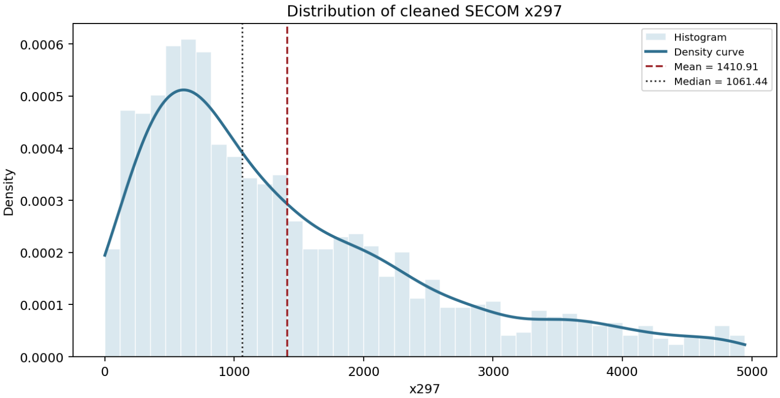
Variable

Selected process variable: x_{297}

- Reason: Low missing, High distinct
- Samples: 1567
- Missing samples removed: 2
- Outlier: 1.5 IQR filter after dropping missing values
- Outliers removed: 128
- Cleaned samples: **1437**

Variable Summary

Mean	SD	Min	Q1	Median	Q3	Max
1410.91	1126.51	0	556.99	1061.44	1995.39	4945.92



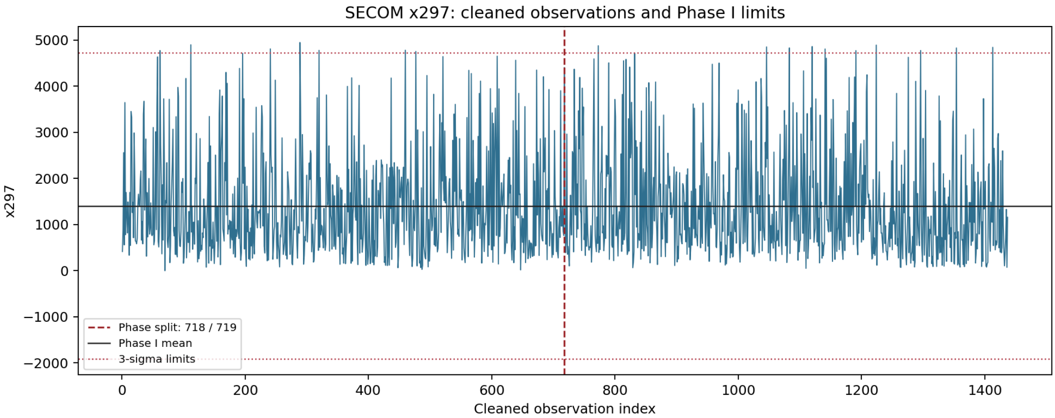
Phase I / Phase II Split

$$m = \lfloor 0.5n \rfloor = \lfloor 0.5(1437) \rfloor = 718$$

Phase	samples	Mean / SD
Phase I	718	1403.64 / 1105.53
Phase II	719	1418.18 / 1147.80

- Phase I is used as the in-control reference sample
- Phase II is monitored sequentially using the chart limits estimated from Phase I

Cleaned Series and Phase Split



Phase I RS/P Diagnosis

Recursive segmentation + permutation result

$$p_{\text{level}} = 0.8675, \quad p_{\text{scale}} = 0.8325$$

- Phase I shows no signifacant evidence of level or scale change
- Phase I data can be regarded as stable

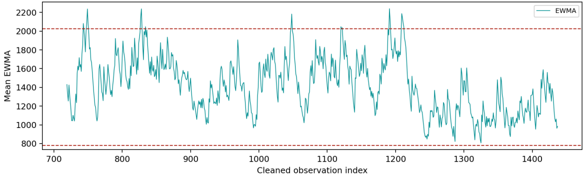
Chart Design

Method	Target	Main setting
Individuals Shewhart	Mean	3-sigma limits
Mean EWMA	Mean	$\lambda = 0.1,\ \rho = 2.454$
Variance EWMA	Variance	$\lambda = 0.1,\ \rho_U = 2.595,\ \rho_L = 1.580$
Adaptive EWMA	Mean	eta1 case (ii), $h = 0.7874$
Mean CUSUM	Mean	$k = 0.5,\ h = 4.095$
Variance CUSUM	Variance	$k_U = 1.848,\ h_U = 7.416;\ k_L = 0.462,\ h_L = -2.446$
CPD charts	Mean / variance / joint	$\alpha = 0.005$

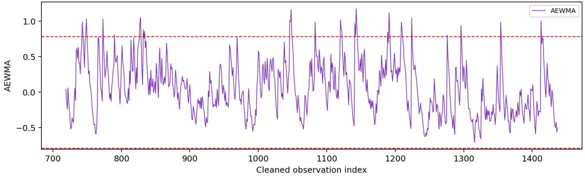
- EWMA and CUSUM settings follow the in-class R code.
- CPD limits follow Chapter 6 approximations for mean, variance, and joint charts.

Mean-Shift Monitoring Results (Main Mean Charts)

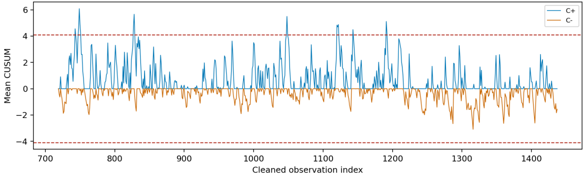
Mean EWMA
First signal: 743
Signal count: 17



Adaptive EWMA
First signal: 743
Signal count: 32



Mean CUSUM
First signal: 743
Signal count: 13

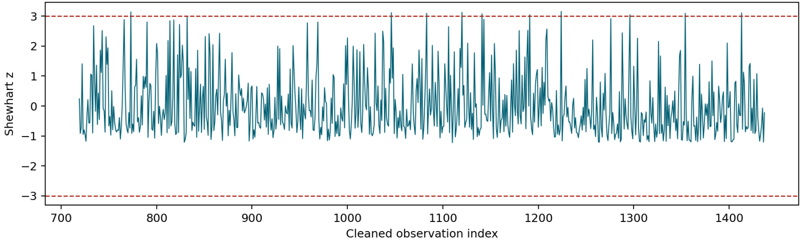


Mean-Shift Monitoring Results (Supplementary Charts)

Individuals Shewhart

First signal: 773

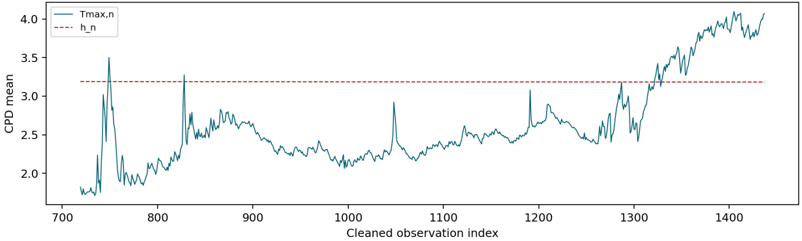
Signal count: 10



CPD Mean

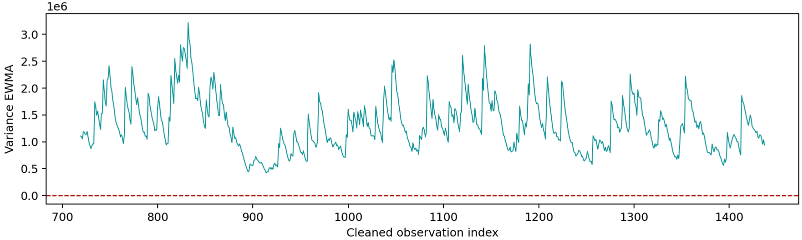
First signal: 749

Signal count: 118

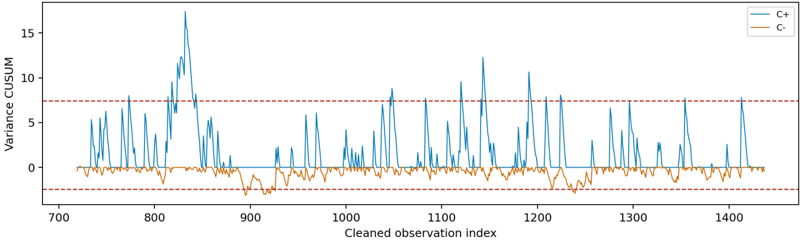


Variance-Shift Monitoring Results (Target: Variance)

Variance EWMA
First signal: 719
Signal count: 719

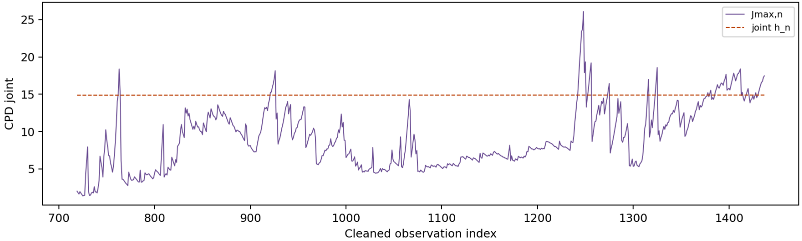


Variance CUSUM
First signal: 773
Signal count: 59

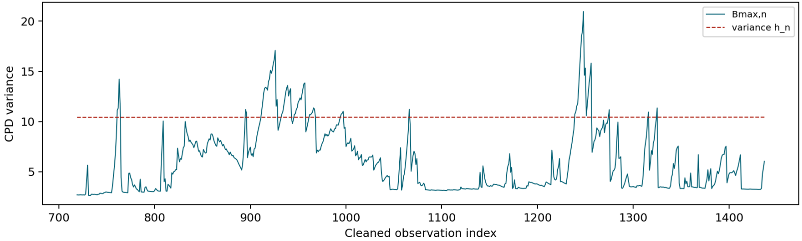


CPD and Joint Monitoring Results

CPD joint
First signal: 762
Signal count: 71



CPD variance
First signal: 761
Signal count: 78



CPD Post-Signal Interpretation

- The strongest CPD variance and joint signals estimate the likely change location around:

$$\widehat{r} \approx 1229.$$

- The strongest CPD mean signal estimates:

$$\widehat{r} \approx 1212$$

Interpretation

- Phase I RS/P diagnosis does not reject stability, so the reference sample is usable.
- Phase II mean changes are detected by EWMA, AEWMA, CUSUM, and CPD mean chart.
- Variance-sensitive methods are more aggressive, especially variance EWMA.
- CPD variance and joint charts locate later structural changes around indices 1229.
- Overall pattern suggests not a single isolated mean shift only, but a combination of mean instability and variance/structural change.

Conclusion

1. The selected SECOM variable x_{297} is usable after missing-value removal and IQR outlier filtering.
2. Phase I RS/P results support using the first 718 cleaned samples as the IC reference.
3. The earliest signal is variance EWMA at the first Phase II observation.
4. For mean-shift detection, mean EWMA, AEWMA, and mean CUSUM signal earliest at index 743.
5. CPD charts provide additional localization and suggest later structural changes around indices 1212–1229.